Expert Systems with Applications 38 (2011) 564–570

Contents lists available at ScienceDirect

Expert Systems with Applications

journal homepage: www.elsevier.com/locate/eswa

A DEA-inspired procedure for the aggregation of preferences

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ARTICLE INFO

Keywords:
Group decision-making
Data Envelopment Analysis
Goal programming

ABSTRACT

Among the most commonly applied methods for the aggregation of individual preferences, weighted scoring rules associate each alternative with a weighted sum of votes received and then rank them in terms of this aggregate value. Some authors have argued that it is not possible to fairly evaluate a set of alternatives by only considering an externally imposed weight scheme. For this reason, researchers have developed certain procedures in which the weights associated with the votes become variables in the model. Data Envelopment Analysis (DEA) represents one class of such models.

In this paper, I propose a new preference-aggregation procedure. The procedure maintains the philosophy inherent in DEA, allowing each alternative to have its own vector of weights, but also introduces a new objective in the evaluation, the optimisation of the rank position in which the alternative is placed, and to avoid the problem of diverse weights by determining a social ranking that uses a common vector of weights.

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1. Introduction

The manner in which a group or social ranking from a set of individual preferences is achieved is an important aspect of the decision-making context. This class of procedures contains either those in which each decision maker (DM) expresses his or her preferences in terms of a rank order of alternatives or those in which each DM is asked to select a subset of alternatives from a feasible set and then rank them from least to most preferred. Among the procedures developed for these situations, one well-known family of methods is weighted scoring rules.

Weighting scoring rules operate by computing, for each alternative, a score that depends on the rank position of the alternative in the individual's order of preferences. Subsequently, the alternatives are ranked by the sum of scores received. The value obtained by the ith alternative is computed as $V_i = \sum_j w_i v_{ij}$, where v_{ij} denotes the number of jth-place ranks that alternative i occupies and w_j is the scoring or weighting applied to the rank-position votes. Because a number is assigned to each candidate, these procedures guarantee a weak ordering of alternatives.

The Borda–Kendall rule is considered the origin of this class of preference-aggregation procedures. Among the most widely discussed scoring rules are (leaving the Borda–Kendall rule aside) the plurality rule, the antiplurality rule and other collective decision-making procedures inspired by the Borda–Kendall method, such as the proposals of Nanson and Copeland (see Fishburn, 1977). For a complete revision of this class of procedures, see, the work of Chebotarev and Shamis (1998), among others.

In real applications, however, it may be desirable to obtain an evaluation of alternatives without externally specifying the weights associated with the votes. Cook and Kress (1990) state that models involving an imposed set of weights fail to provide a fair overall assessment. Each evaluation that implies the use of an externally imposed scoring vector is somewhat arbitrary. Note that a non-winning alternative with the vector considered for the evaluation could be a winner if another weighting vector is applied.

In addition, in group decision problems in which multiple DMs are involved, it is easier to achieve agreement with respect to a set of feasible weights than with respect to a unique vector (as traditional scoring rules require). In this case, reaching a consensus requires determining the minimum information set that contains the preference relations elicited by the DMs, expressed in terms of bounds or inequalities in the weighting vector's components. These two issues help derive the models in which the weights (or scores) associated with each rank position are variables in the problem instead of models that evaluate alternatives with an a priori scoring vector.

Among this class of procedures, I highlight those that are based on Data Envelopment Analysis (DEA). Cook and Kress (1990) propose the initial preference-aggregation model based on a DEA methodology to avoid subjectivity in the choice of a vector of weights. The model evaluates each alternative according to its most favourable scoring vector. Following the philosophy of DEA, each alternative is allowed its own vector of weights to be evaluated. The authors proposes a DEA/AR model, in which the number of votes received at a particular rank position represents an output,

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doi:10.1016/j.eswa.2010.07.002

and the model considers a unique input equal to unity. For the weights, some constraints assure that a better rank position yields a larger weight when included in the model. This set of constraints is denoted in DEA terminology as an assurance region (AR).

With respect to DEA-based models, two main criticisms appear in the literature: multiple top-ties and overly diverse weights. DEA models use assignments of the same aggregate value (equal to unity) to evaluate multiple alternatives as efficient. There is no criterion to discriminate among these alternatives in order to construct a ranking of alternatives. Hence, this approach assumes a drawback to those procedures in which the goal is to construct a rank order of alternatives.

On the other hand, in this procedure, overly diverse weights can appear, given that each alternative can have its own vector of weights (i.e., the one that maximises its aggregate value). The social ranking, which uses different weights for various alternatives, might be rather unacceptable because the traditional ranking uses common weights across alternatives (see Hashimoto & Wu, 2003 for a detailed discussion of this aspect of the approach).

To resolve these two tasks, alternative models have been developed. In Adler, Friedmand, and Sinvany-Stern (2002), the authors summarise the methods proposed for ranking alternatives in the DEA context, yielding a solution to the problem of multiple top-ties and then classifying them into six categories. In the case of a preference-aggregation problem, certain DEA-based procedures have been discussed that, in different ways, attempt to avoid the aforementioned drawback. A complete review of the main models can be seen, for instance, in Llamazares and Pea (2009).

The initial model developed by Cook and Kress (1990) assures that better rank positions yield larger weights by including a discriminating intensity function. In order to construct a ranking of alternatives, the proposed model maximises the discriminating factor. A discrimination procedure based on the cross-evaluation is proposed in Green, Doyle, and Cook (1996) and in Sexton, Silkman, and Hogan (1986) to avoid the problem of choosing a discrimination function that appears in the previous procedure. In Obata and Ishii (2003), the authors demonstrate that in order to compare the aggregate values received by alternatives, it is first necessary to use a weighting vector with the same size. For this reason, they propose normalising the vectors of weights.

In order to discriminate among efficient alternatives, Hashimoto (1997) addresses an AR/exclusion model based on the concept of super-efficiency proposed in Andersen and Petersen (1993). To avoid the problem of diverse weights, the author proposes in Hashimoto and Wu (2003) a model that results in a common vector of weights for evaluating the set of proposed alternatives. More recently, the work of Zerafat Angiz, Emrouznejad, Mustafa, and Rashidi Komijan (2009) introduces a new mathematical method, inspired by DEA methodology, which determines the importance of rank positions according to decision makers in order to reach a more realistic solution.

In this paper, I propose a new model inspired by DEA methodology. The procedure does not specify an a priori vector and consists of two stages. First, a DEA-inspired model for the aggregation of preferences is applied, wherein the objective is not the maximisation of the aggregated value but rather the ordinal position induced by these values. Second, in order to obtain a group solution, the procedure derives a compromise solution by determining a social vector of weights for evaluating the complete set of alternatives.

The rest of the paper is organised as follows. In Section 2, the proposed procedure is described, including the derivation of a linear model to determine the solution in both stages. In Section 3, the model is adapted to a case in which ties between alternatives are permitted, generating weak orders in both stages. Section 4 includes an illustrative example, and Section 5 draws conclusions.

2. Aggregating preferences with a DEA-inspired procedure

Let $X = \{x_1, \ldots, x_N\}$ be a set of N ($N \geqslant 3$) alternatives that have been evaluated by a group of decision makers. Each decisor gives preferences by selecting a subset of K alternatives (or the complete set X) and ranking them from most to least preferred. An order vector can then be derived from the preferences of each DM, such as a vector that contains the names of alternatives in a preference sequence.

From each order vector, a priority or rank vector can be constructed by assigning the position number within the order vector, where the relative position of each alternative is represented in the corresponding order. By convention, the value 1 is assigned to the most important alternative and N to the least important. Note that a priority vector contains the rank values of objects, whereas an order vector contains the names of alternatives. In this paper, rank values will be used for computational purposes, and object names will be used for descriptive purposes.

The aggregation-of-preferences models based on DEA attempt to maximise the aggregate value of votes obtained by each alternative. For each individual alternative, I estimate a maximisation model that gives each alternative the opportunity to have its own scoring vector.

The main idea can be summarised in the following model,

$$\begin{aligned} V^o(x_o) = \max \quad & \sum_{j=1}^{K} w_j^o v_{oj}, \\ \text{s.t.} \quad & \sum_{j=1}^{K} w_j^o v_{ij} \leqslant 1, \quad i = 1, \ldots, N, \\ & w_j^o - w_{j+1}^o \geqslant d(j, \varepsilon), \quad j = 1, \ldots, K-1, \\ & w_K^o \geqslant d(K, \varepsilon). \end{aligned} \tag{1}$$

Model (1) corresponds to the evaluation of alternative x_o. The term w_j^o denotes the weight associated with the jth-rank-position votes in the evaluation of alternative x_o, v_{ij} represents the number of the jth-rank positions votes obtained by alternative x_i, and $V^o(x_i) = \sum_{j=1}^{K} w_j^o v_{ij}$ represents the aggregate value associated with alternative x_i when the optimal weighting vector of alternative x_o is considered. The term $d(j, \varepsilon)$ denotes the discrimination intensity function, where ε represents a non-Archimedean element. This set of constraints implies that the ranking of alternatives has been constructed from most to least preferred, and therefore, the weight assigned to the votes must reflect this feature.

In this context, I propose a two-stage model for determining a group ranking of alternatives. The first stage develops a DEA-inspired model but with a different objective. For each alternative, the model tries to determine a vector of weights that verifies the objective of optimising the rank position of each alternative. That is, the model looks for the weighting vector that minimises the rank position that the alternative occupies in the ranking of alternatives induced by the aggregate values $V^o(x_i)$. The second stage determines a compromise solution for the group after considering the values obtained by each alternative in the previous step, according to each alternative's aspiration levels.

2.1. First stage: optimisation of the ranking position

In the first stage, each alternative is evaluated individually, similar to model (1) but with a different objective. Taking into account the fact that the target of this class of procedures is to construct a ranking of alternatives and the fact that the numeric values of $V^o(x_i)$ are used only for comparative purposes, it is more appropriate to minimise the rank position to describe what the best attainable situation for an alternative is.

In each evaluation model, a complete order for alternatives can be induced from the aggregated values, $V^o(x_i)$, given that the alternative x_i is preferred to x_h with the vector w^o if $V^o(x_i) > V^o(x_h)$. In this case, the objective is to construct a linear order of alternatives, and ties between them are not permitted. For those cases in which the aggregate values of two or more alternatives are the same, the model selects the most favourable rank position for the alternative under evaluation.

It is easy to see that both evaluations coincide for *efficient* alternatives. Obviously, if one alternative can achieve the maximum value in model (1) (i.e., $V^o(x_o) = 1$), this alternative will be ranked at the top of the order. However, in the case of *inefficient* alternatives, the results are not necessarily the same when the objective changes from the maximisation of $V^o(x_i)$ to the minimisation of the rank position. For the second objective, the alternative x_o can make its aggregate value worse if this implies achieving a better position in the order of alternatives as a consequence of worse values or any additional alternatives.

The term $R^o = (r_1^o, \ldots, r_N^o)$ denotes the priority vector induced in the evaluation of alternatives x_o, where r_i^o represents the rank position given to the ith alternative. To determine the best rank position for x_o, as denoted by r_o^o, the following mixed-integer programming model is solved,

$$\begin{aligned}
\min \quad & r_o^o \\
\text{s.t.} \quad & \sum_{j=1}^{k} w_j^o v_{ij} - \sum_{j=1}^{k} w_j^o v_{hj} + \delta_{ih}^o M \geqslant 0, \quad i \neq h, \\
& \delta_{ih}^o + \delta_{hi}^o = 1, \quad i \neq h, \\
& \delta_{ih}^o + \delta_{hk}^o + \delta_{ki}^o > 1, \quad i \neq h \neq k, \\
& r_i^o = 1 + \sum_{i \neq h} \delta_{ih}^o, \quad i = 1, \ldots, N, \\
& w^o \in \Phi, \\
& \delta_{ih}^o \in \{0, 1\},
\end{aligned} \tag{2}$$

where M is a large positive quantity and Φ specifies the plausible conditions for the weights.

Set Φ should contain, at minimum, a set of constraints such that $\{w^o \in \mathbb{R}^K,\ \sum_{j=1}^{K} w_j^o = 1,\ w_1^o \geqslant w_2^o \geqslant \cdots \geqslant w_K^o \geqslant 0\}$. Any additional information about the discrimination between weights associated with rank positions can be included in set Φ by incorporating the corresponding set of constraints. The interpretation of Φ is as follows: every vector w^o included in the partial information set Φ is considered reasonable by the DMs, because they are all in agreement as to the information about the weights, although each vector can induce a different weak order of alternatives through the values $V^o(x_i)$. In the case in which Φ contains a unique vector (i.e., the case of traditional models with an a priori defined weighting vector), the problem is solved, and the final ranking can be constructed by comparing the aggregated values of the alternatives. In general, however, the set Φ contains multiple vectors, and hence, it becomes necessary to establish a criterion to determine which vector should be selected for evaluating alternatives. In this case, for each alternative, the vector that minimises its rank position is selected.

It is worth noting that the normalisation condition $\sum_{j=1}^{K} w_j^o = 1$ has been included in order to substitute for the traditional DEA restriction that guarantees that all aggregated values are less than or equal to unity. This normalisation condition permits a superior interpretation of the weights in terms of the relative importance associated with the votes in different positions and permits the inclusion of simpler numeric bounds to the values of the weights.

Model (2) computes pairwise comparisons of the alternatives and constructs the corresponding linear order that these values induce. Note that the binary variable δ_{ih}^o should be equal to unity if x_i is strictly dominated by x_h, i.e., whenever $\sum_{j=1}^{k} w_j^o v_{ij} < \sum_{j=1}^{k} w_j^o v_{hj}$. Thus, $\delta_{ih}^o = 1$ implies that alternative x_h must be ranked ahead of x_i. Consequently, the rank position of x_i is the sum of variables δ_{ih}^o plus one (note that for the alternative ranked at the first position, the corresponding variables are null). The second and third groups of constraints guarantee that a linear order (without ties) is obtained as the solution. The constraints such as $\delta_{ih}^o + \delta_{hi}^o = 1$ assure that either x_i is preferred to x_h or x_h is preferred to x_i.

It worth noting that, in the case of a tie, $\sum_{j=1}^{k} w_j^o v_{ij} = \sum_{j=1}^{k} w_j^o v_{hj}$, the binary variables δ_{ih}^o and δ_{hi}^o can indistinctly take the value one or be null, and in every case, the constraints are verified. The selection of which variable will be equal to one or zero is performed in order to elicit the rank position r_o^o that is most favourable for the alternative x_o. The constraints such as $\delta_{ih}^o + \delta_{hk}^o + \delta_{ki}^o > 1$ are included in order to adequately represent ties between more than two alternatives.

Note that the solution to (2) includes an order of alternatives and a weighting vector w^o, both of which are selected to optimise the position of alternative x_o in the linear order induced by the aggregate values of the votes.

2.2. Second stage: a common-weights group solution

One of the main drawbacks of DEA-based models is that overly diverse weights may appear because each alternative can have its own vector of weights to be evaluated. This characteristic may be rather unacceptable for using a common vector of weights across alternatives. For this reason, I determine a common-weights ranking of alternatives in the second stage of the procedure.

Consider that the result of the previous stage is summarised in the vector $R^I = (r_1^I, \ldots, r_N^I)$, where $r_i^I = r_i^i$. Each element of R^I represents the best position that each alternative can aspire to, considering the vectors of weights included in the set Φ. Therefore, these values represent an ideal solution that is very often impossible to satisfy. The second stage proposes a model that selects a common vector of weights (from those included in Φ) that minimises the distance to the aspiration level of each alternative, represented by its corresponding value r_i^i.

In order to determine the group order, the following mixed-integer linear model is solved,

$$\begin{aligned}
\min \quad & \sum_{i=1}^{N} \alpha_i \\
\text{s.t.} \quad & \sum_{j=1}^{k} w_j v_{ij} - \sum_{j=1}^{k} w_j v_{hj} + \delta_{ih} M \geqslant 0, \quad i \neq h, \\
& \delta_{ih} + \delta_{hi} = 1, \quad i \neq h, \\
& \delta_{ih} + \delta_{hk} + \delta_{ki} > 1, \quad i \neq h \neq k, \\
& r_i^G = 1 + \sum_{i \neq h} \delta_{ih}, \quad i = 1 \ldots, N, \\
& r_i^G - \alpha_i = r_i^i, \quad i = 1, \ldots, N, \\
& w \in \Phi, \\
& \delta_{ih} \in \{0, 1\} \quad i \neq h.
\end{aligned} \tag{3}$$

Some aspects of model (3) must be highlighted. First, the solution to the model is a linear order of alternatives, $R^G = (r_1^G, \ldots, r_N^G)$. This rank order is induced by the aggregate values of the votes, very similar to model (2). By selecting an appropriate vector from the set Φ, the model constructs the order of alternatives such that the distance to the values offered by each alternative is minimised. Hence, a social evaluation has been made using a common vector of weights, but considering the most achievable position for each alternative through the aspiration levels, from all of the vectors contained in set Φ.

Model (3)'s objective is to minimise the total distance across alternatives. Some alternative objective could also be computed. I

propose the minimisation of the maximum distance across alternatives. In this case, the resultant model is

$$\begin{aligned}
\min \quad & D \\
\text{s.t.} \quad & \sum_{j=1}^{k} w_j v_{ij} - \sum_{j=1}^{k} w_j v_{hj} + \delta_{ih} M \geqslant 0, \quad i \neq h, \\
& \delta_{ih} + \delta_{hi} = 1, \quad i \neq h, \\
& \delta_{ih} + \delta_{hk} + \delta_{ki} > 1, \quad i \neq h, \\
& r_i^G = 1 + \sum_{i \neq h} \delta_{ih}, \quad i = 1, \ldots, N, \\
& r_i^G - \alpha_i = r_i^i, \quad i = 1, \ldots, N, \\
& D > \alpha_i, \quad i = 1, \ldots, N, \\
& w \in \Phi, \\
& \delta_{ih} \in \{0, 1\} \quad i \neq h.
\end{aligned} \tag{4}$$

Variable D computes the maximum distance across alternatives. This model seems more appropriate for this context because the first stage proposes an individual evaluation of alternatives. Therefore, it seems adequate in this second step to individually consider the alternatives in order to determine the group order.

The procedure offers a compromise solution and so verifies some of the properties desirable in the context of social choice processes. The properties of feasibility, anonymity, neutrality and non-dictatorship are direct consequences of the construction of this procedure.

Feasibility guarantees that the group's solution is always obtainable. Verification of the property of anonymity ensures that the procedure treats all group members in the same way, because the collective ranking depends only on the number of DMs in each individual evaluation. With the property of neutrality, all alternatives are treated equally. More formally, any permutation in the set of alternatives induces a corresponding permutation in the consensus ordinal ranking. The non-dictatorship property implies that no single DM's individual preferences can determine the consensus ranking.

3. Constructing weak orders

In the previous section, the orders induced by the votes' aggregate values do not represent the situation in which two or more alternatives obtain the same value. In this case, the alternative under evaluation is ranked ahead of the tied alternatives. The goal of this section is to obtain a model that more accurately reflects this situation by inducing weak orders from the aggregate values.

In this context, the model requires some way of determining the weak orders induced by each vector $w^o \in \Phi$ (though the values $V^o(x_i)$). Note that this procedure must not only induce the rank positions from aggregated values, as in the previous procedure, but must also adequately represent ties between alternatives. For this purpose, a priority vector will be constructed for each order. In the case of a tie, the average of the values corresponding to tied alternatives is assigned.

Considering the procedure developed in Contreras (2010), the values for the corresponding priority vector are determined by means of the binary variables σ_{ih}^o and γ_{ih}^o (with $i, h = 1, \ldots, N$). Consider the following set of constraints,

Table 1
Values of binary variables.

	σ_{ih}^o	σ_{hi}^o	γ_{ih}^o	γ_{hi}^o
$V^o(x_i) > V^o(x_h)$	0	1	0	1
$V^o(x_i) = V^o(x_h)$	0	0	1	1
$V^o(x_i) < V^o(x_h)$	1	0	1	0

$$\begin{aligned}
& V^o(x_i) - V^o(x_h) + \sigma_{ih}^o M \geqslant 0, \quad \forall i \neq h, \\
& V^o(x_i) - V^o(x_h) + \gamma_{ih}^o M \geqslant \varphi, \quad \forall i \neq h;
\end{aligned} \tag{5}$$

where M denotes a large number and φ is the discriminating factor between alternatives included for computational purposes. That is, we say that alternative x_i strictly dominates x_h in the evaluation of x_o if $V^o(x_i) - V^o(x_h) \geqslant \varphi$.

The variable σ_{ih}^o should be equal to unity if x_i is strictly dominated by x_h, i.e., whenever $V^o(x_h) > V^o(x_i)$. In contrast, γ_{ih}^o should be one whenever $V^o(x_h) \geqslant V^o(x_i)$. This variable is used to represent ties between alternatives. Table 1 summarises the correct values that the binary variables should adopt in each case.

To guarantee that the values' assignments are carried out in the correct way, we must include the following sets of constraints,

$$\sigma_{ih}^o + \sigma_{hi}^o \leqslant 1, \forall i \neq h; \tag{6}$$

$$\sigma_{ih}^o + \gamma_{hi}^o = 1, \forall i \neq h. \tag{7}$$

These constraints are necessary in order to induce a weak order of alternatives from the values $V^o(x_i)$. These equations guarantee that the assignment of values to the binary variables σ_{ih}^o and γ_{ih}^o is correct in the sense that, only if $V^o(x_h) > V^o(x_i)$, then $\sigma_{ih}^o = 1$ and consequently $\sigma_{hi}^o = 0$. The values of γ_{ih}^o will also depend on the value assigned to σ_{ih}^o, and these values represent ties between alternatives. It is worth bearing in mind that, in each constraint, we must consider not only the dominance relation of x_i over x_h but also the relation of x_h to x_i.

The position of the alternative x_i in the priority vector R^o (which represents the order induced by the evaluation of the alternative x_o) can be obtained as the sum of the binary variables σ_{ih}^o and γ_{ih}^o over 2, plus one. That is,

$$r_i^o = \sum_{i \neq h} \frac{\sigma_{ih}^o + \gamma_{ih}^o}{2} + 1. \tag{8}$$

This expression adds unity for each alternative ranked above x_i (in this case, both variables, σ_{ih}^o and γ_{ih}^o, are equal to one) and 0.5 for each alternative that ties with x_i (in that case, σ_{ih}^o is equal to zero and γ_{ih}^o is equal to one). To determine the value of r_i^o, unity is added to the total sum.

The constraint in (9), together with the other constraints described above, assures that the vector R^o represents a weak order of alternatives in the standard form (assigning the average of rank positions to the tied alternatives),

$$\sum_{i=1}^{N} r_i^o = \frac{N(N+1)}{2}. \tag{9}$$

The model that evaluates the alternative x_o in the best situation becomes

$$\begin{aligned}
\min \quad & r_o^o \\
\text{s.t.} \quad & \sum_{j=1}^{k} w_j^o v_{ij} - \sum_{j=1}^{k} w_j^o v_{hj} + \sigma_{ih}^o M \geqslant 0, \quad i \neq h, \\
& \sum_{j=1}^{k} w_j^o v_{ij} - \sum_{j=1}^{k} w_j^o v_{hj} + \gamma_{ih}^o M \geqslant \varphi, \quad i \neq h, \\
& \sigma_{ih}^o + \sigma_{hi}^o \leqslant 1, \quad i \neq h, \\
& \sigma_{ih}^o + \gamma_{hi}^o = 1, \quad i \neq h, \\
& r_i^o = 1 + \sum_{i \neq h} \frac{\sigma_{ih}^o + \gamma_{ih}^o}{2} + 1, \quad i = 1, \ldots, N, \\
& \sum_{i=1}^{N} r_i^o = \frac{N(N+1)}{2}, \\
& w^o \in \Phi, \\
& \sigma_{ih}^o, \gamma_{ih}^o \in \{0, 1\}.
\end{aligned} \tag{10}$$

Whereas the results of model (2) are integer values, model (10) gives the average of two or more rank positions in the case of ties. The solution minimises r_o^o by choosing a weighting vector $w \in \Phi$, taking into account the fact that, if one or more alternatives achieve the aggregate value $V^o(x_o)$, this can lead to a worse result for x_o.

In the second stage, which obtains a common-weight solution R^G, the model minimising the total disagreement is as follows,

$$\begin{aligned}
\min \quad & \sum_{i=1}^{N} \alpha_i \\
\text{s.t.} \quad & \sum_{j=1}^{k} w_j v_{ij} - \sum_{j=1}^{k} w_j v_{hj} + \sigma_{ih} M \geqslant 0, \quad i \neq h, \\
& \sum_{j=1}^{k} w_j v_{ij} - \sum_{j=1}^{k} w_j v_{hj} + \gamma_{ih} M \geqslant \varphi, \quad i \neq h, \\
& \sigma_{ih} + \sigma_{hi} \leqslant 1, \quad i \neq h, \\
& \sigma_{ih} + \gamma_{hi} = 1, \quad i \neq h, \\
& r_i^G = 1 + \sum_{i \neq h} \frac{\sigma_{ih} + \gamma_{ih}}{2} + 1, \quad i = 1, \ldots, N, \\
& \sum_{i=1}^{N} r_i^* = \frac{N(N+1)}{2}, \\
& r_i^G - \alpha_i = r_i^i, \quad i = 1, \ldots, N, \\
& w \in \Phi, \\
& \sigma_{ih}, \gamma_{ih} \in \{0, 1\}.
\end{aligned} \tag{11}$$

The model that minimises the maximum disagreement be constructed similarly to model (4), by including the objective of minimising the variable D, along with a set of constraints such that $D > \alpha_i$ for all $i = 1, \ldots, N$.

This extension of the model developed in the previous section verifies the previously indicated properties in the social choice context.

4. Illustrative example

Consider the following example, taken from Green et al. (1996), in which a group of decision makers must decide among a set of seven alternatives denoted by $X = \{x_1, \ldots, x_7\}$. Each agent is required to select and rank the best two alternatives. The results obtained for each decision maker are summarised in Table 2.

Table 2
Votes received in each position.

Alternative	First	Second
x_1	32	10
x_2	28	20
x_3	13	36
x_4	20	27
x_5	27	19
x_6	30	8
x_7	0	30

Suppose that all DMs are in agreement as to representing the relative importance of the rank position by means of the set of feasible weighting vectors, $\Phi = \{w_1 \geqslant 3w_2 \geqslant 0.03, w_1 + w_2 = 1\}$. This set can be interpreted to mean that the one vote received in the first position is valued as at least three times the value of a vote received in the second position. In addition, a lower bound of 0.01 is imposed onto the weight of the second position votes (to assure that all are considered in the evaluation). A normalisation condition is included such that the sum of the weighting vector's components is equal to unity.

To determine the social solution first, I proceed to solve model (2). The results are summarised in Table 3, which includes the aggregate values and the rankings induced in the evaluation of each alternative. Table 4 summarises the weighting vectors used in each evaluation.

Therefore, the aspiration levels for the alternatives at the second stage are $R^I = (1, 2, 6, 5, 3, 2, 7)$.

The solution differs when the objective is maximising the aggregate value of the alternative. The results obtained for the alternative x_5 when the objective is maximising the aggregate value are summarised in Table 5. It is interesting to note that the alternative x_5, which can achieve the first position with the vector $w^5 = (0.750, 0.250)$, results in a larger aggregate value but a worse rank position when using the vector $w^2 = (0.990, 0.010)$.

The second stage consists of determining a compromise solution. Each alternative presents the result of the previous stage as an aspiration level, and the objective is to determine a vector of weights, from those included in set Φ, such that the rank order induced by the aggregate values is closest to the individual requirements. In this case, the minimax solution is determined.

The solution in model (4) is equal to unity. This means that the difference between the rank assigned to the alternatives in the group order and the best rank this alternative could achieve with the weights in Φ is zero (this is the case for the alternatives x_1, x_2, x_3, x_4 and x_7) or one (alternatives x_5 and x_6).

The results for the group are included in Table 6. The vector obtained as the solution is $w = (0.786, 0.214)$, which induces a ranking in which alternative x_1 is placed at the first position, followed by x_2, whereas alternative x_7 occupies the last position. Note that, despite the aggregate values of x_5 and x_6 being the same, the model ranks x_6 ahead of x_5 in order to minimise the differences with the ideal values of R^I.

The model that induces weak orders yields the same values for the vector R^I, although some of the orders induced in the individual evaluations vary. For instance, the order induced in the evaluation of x_1 becomes $R^1 = (1, 2.5, 6, 5, 4, 2.5, 7)$, although these variations do not affect the values of R^I.

Nevertheless, the result of the second stage is different. Note that the result summarised in Table 6 implies that the aggregate values of x_5 and x_6 are the same, although the order induced does not reflect this situation and ranks x_5 ahead of x_6 (in order to assure that the differences with respect to the values of R^I are not greater

Table 3
First stage: Aggregated values and ranking induced.

	$V^1(x_i)$	r_i^1	$V^2(x_i)$	r_i^2	$V^3(x_i)$	r_i^3	$V^4(x_i)$	r_i^4	$V^5(x_i)$	r_i^5	$V^6(x_i)$	r_i^6	$V^7(x_i)$	r_i^7
x_1	28,86	**1**	26,50	1	26,50	1	26,50	1	26,50	1	31,78	1	26,50	1
x_2	26,86	2	26,00	**2**	26,00	2	26,00	2	26,00	2	27,92	3	26,00	2
x_3	16,29	6	18,75	6	18,75	**6**	18,75	6	18,75	6	13,23	6	18,75	6
x_4	21,00	5	21,75	5	21,75	5	21,75	**5**	21,75	5	20,07	5	21,75	5
x_5	25,86	4	25,00	3	25,00	3	25,00	3	25,00	**3**	26,92	4	25,00	3
x_6	26,86	3	24,50	4	24,50	4	24,50	4	24,50	4	29,78	**2**	24,50	4
x_7	4,29	7	7,50	7	7,50	7	7,50	7	7,50	7	0,30	7	7,50	**7**

Table 4
First stage: weighting vectors.

Alternative	w_1^i	w_2^i
x_1	0.857	0.143
x_2	0.750	0.250
x_3	0.750	0.250
x_4	0.750	0.250
x_5	0.750	0.250
x_6	0.990	0.010
x_7	0.750	0.250

Table 5
Evaluation of x_5. Maximization of the aggregate value.

	$V^5(x_i)$	r_i^5
x_1	31,78	1
x_2	27,92	3
x_3	13,23	6
x_4	20,07	5
x_5	**26,92**	**4**
x_6	29,78	2
x_7	0,30	7

Table 6
Second stage: group solution.

	$V(x_i)$	r_i^G
x_1	27,29	1
x_2	26,29	2
x_3	17,93	6
x_4	21,50	5
x_5	25,29	4
x_6	25,29	3
x_7	6,43	7

than unity). If this tie is represented in the standard form, both alternatives are represented in the priority vector by $r_5^G = r_6^G = 3.5$, and the difference between x_5 and its aspiration level ($r_5^I = 2$) is 1.5 units. In order to minimise the maximum difference, the solution is $R^G = (1,3,6,5,4,2,7)$, using the weighting vector $w = (0.990, 0.010)$. In this case, the use of the model based on weak orders is useful for refining model (4)'s initially given solution. Note that both solutions have a maximum distance from R^I of unity. The second solution, however, more accurately reflects the aggregate values obtained by the alternatives.

5. Concluding remarks

The aggregation of individual preferences has wide applications in social choice and voting systems. Some of the most widely used methods in these fields are based on the determination of an aggregated value for each alternative. In these procedures, a score or weight is associated with each rank position, and individual opinions are aggregated by means of a weighted sum of individual scores.

The method for determining the weights associated with rank positions is an important issue because it determines a group's solution. Some authors have argued that an externally imposed scheme of weights does not yield a fair evaluation of alternatives, and these authors have proposed models in which the weights associated with vote types are variables in the models. Following this idea, the models inspired by Data Envelopment Analysis (DEA) give each alternative the opportunity to have its own weighting vector in order to maximise the aggregate value.

In this paper, a two-stage procedure for aggregating individual preferences, as expressed in terms of rank orders, is developed. The procedure preserves the main idea of that based on DEA, where each alternative is determined by its best attainable evaluation, but it also introduces some interesting innovations. First, in the individual evaluations, each alternative optimises the rank position, not the aggregate value. In this context, in which the goal is to determine a ranking of alternatives (and aggregate values are dealt with exclusively to make comparisons between alternatives), the best attainable situation for an alterative must be considered the best rank position in which it can be placed, independent of its associated numeric value.

The classical normalisation constraints of DEA, which establish that all aggregate values must be less than or equal to unity, are substituted for the condition that assures that the sum of weights is equal to one. This makes the incorporation of additional information about the discrimination between weights easier and results in a simpler interpretation with respect to numeric bounds.

Finally, the second stage determines a group ranking using a common vector of weights. Considering the values in the first stage aspiration levels, the model obtains a vector (from those contained in the feasible set) that induces this ranking of alternatives (through the aggregate values) that is closest to the ideal situation of each one. This second stage permits the consideration of different solutions, such as the minimisation of the total disagreement or the minimisation of the maximum disagreement.

Two possible approaches have been considered in which the treatment of ties between alternatives is different. In the first approach, the models induce a linear order from the aggregate values, and in the case of ties, the tied alternatives are ranked freely in order to minimise the distance to the best attainable position for each one. In the second approach, where weak orders are induced, ties between alternatives are represented in the priority vectors by assigning the average of the corresponding rank positions. This feature can affect the solution, because the model is not free to place those alternatives that obtain the same aggregate value in the order.

This procedure can be easily adapted to the case in which individual preferences are expressed in terms of ranking of a complete set of alternatives or by placing the alternatives on an ordinal scale. In addition, the procedure allows for the inclusion of additional information, such as the inclusion in the second stage of a coefficient that reflects the relative importance of the alternatives in the group decision.

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